

TYPES OF PROMPTS

1. Straight forward Prompts

a) Prompt:

Explain the working principle of a DC motor in simple terms

Response:

Working Principle of a DC Motor (Simple Explanation)

A DC motor works on the basic principle that a current-carrying conductor placed in a magnetic field experiences a force. This idea comes from the scientific concept known as the Lorentz Force.

How it works step by step:

1. When DC supply is given, current flows through the armature (coil).
2. The armature is placed inside a magnetic field created by field magnets.
3. Due to the interaction between the magnetic field and current, a force acts on the conductor.
4. This force causes the armature to rotate.
5. A component called the commutator reverses the current direction, ensuring continuous rotation.

b) Prompt:

List the steps to connect a laptop to a Wi-Fi network.

Response:

Steps to Connect a Laptop to a Wi-Fi Network

1. Turn on Wi-Fi
Make sure the Wi-Fi is enabled on your laptop (using the keyboard shortcut or settings).
2. Open Network Settings
Click on the Wi-Fi icon on the taskbar (usually at the bottom-right corner of the screen).
3. View Available Networks
A list of available Wi-Fi networks will appear.

4. Select Your Network
Choose the desired Wi-Fi network (SSID) from the list.
5. Click Connect
Press the “Connect” button.
6. Enter Password
Type the correct Wi-Fi password (security key).
7. Confirm Connection
Click “Next” or “Connect” and wait for the laptop to establish the connection.
8. Check Connection Status
Once connected, you’ll see a “Connected” message or Wi-Fi symbol.

2. Tabular Format Prompting

a) Prompt:

Compare AC motor and DC motor in tabular form based on working principle, efficiency, maintenance, and applications.

Response:

Comparison of AC Motor and DC Motor

Parameter	AC Motor	DC Motor
Working Principle	Works on alternating current; uses a rotating magnetic field to produce motion	Works on direct current; based on the Lorentz Force (force on current-carrying conductor)
Efficiency	Generally high, especially in induction motors	Slightly lower due to losses in brushes and commutator
Maintenance	Low maintenance (no brushes or commutator)	High maintenance (brushes and commutator wear out)
Applications	Fans, pumps, compressors, industrial machines	Electric vehicles, cranes, elevators, robotics

b) Prompt:

Present the differences between Wi-Fi and mobile data in a tabular format based on speed, cost, availability, and reliability.

Response:

Differences Between Wi-Fi and Mobile Data

Parameter	Wi-Fi	Mobile Data
Speed	Usually faster; depends on router and broadband connection	Varies based on network type (3G/4G/5G); can be slower or unstable at times
Cost	Generally cheaper; fixed monthly broadband plans	More expensive; charged based on data usage or limited plans
Availability	Limited to specific locations (home, office, public hotspots)	Available almost everywhere with network coverage
Reliability	More stable and consistent in a fixed location	Can fluctuate due to signal strength, congestion, or location

3.Preceding Question Prompting

a) Prompt:

What is electromagnetic induction? Based on this concept, explain the working principle of a transformer.

Response:

Working Principle of a Transformer:

A transformer works based on electromagnetic induction.

1. When alternating current (AC) flows through the primary winding, it produces a changing magnetic field.
2. This changing magnetic field creates a varying magnetic flux in the transformer core.
3. The varying flux links with the secondary winding.
4. According to electromagnetic induction, an emf is induced in the secondary winding.
5. This induced emf provides the output voltage.

b) Prompt:

What are the basic requirements for internet connectivity? Based on that, explain how a smartphone connects to the internet.

Response:

Basic Requirements for Internet Connectivity:

1. **Device** – A smartphone, laptop, or computer
 2. **Network Interface** – Hardware like Wi-Fi or mobile data (cellular radio)
 3. **Internet Service Provider (ISP)** – A company such as Airtel or Jio
 4. **Transmission Medium** – Fiber cables, radio waves, or satellite signals
 5. **IP Address** – A unique identifier for communication on the internet
 6. **Protocols** – Rules like TCP/IP that govern data transfer
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How a Smartphone Connects to the Internet:

1. The smartphone enables **Wi-Fi or mobile data**.
2. It connects to a nearby **Wi-Fi router** or **cellular tower**.
3. The network is provided by an ISP (like Airtel or Jio).
4. The ISP assigns an **IP address** to the smartphone.
5. Using protocols like TCP/IP, the phone sends and receives data.
6. Data travels between the smartphone and internet servers, allowing browsing, apps, and communication.

4.Missing Word Prompting

a) Prompt:

A transformer works on the principle of _____ induction.

Response:

A transformer works on the principle of electromagnetic induction.

b) Prompt:

To connect to a Wi-Fi network, you must enter the correct _____.

Response:

To connect to a Wi-Fi network, you must enter the correct password (security key).